

NEPTUNE — Complete 3D Assembly Master

Every component, outside and in, assembled and exploded. Numbers are constant across all figures. Geometry is schematic-proportioned; dimensions live in the integration doc.

FIG A · Boat assembled — external

FIG A — Boat assembled

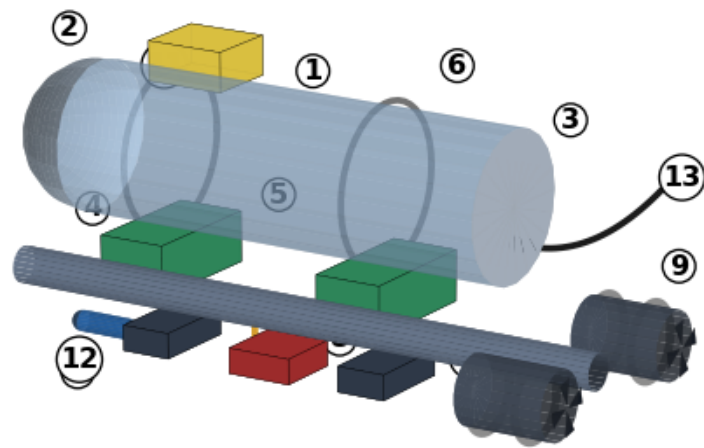
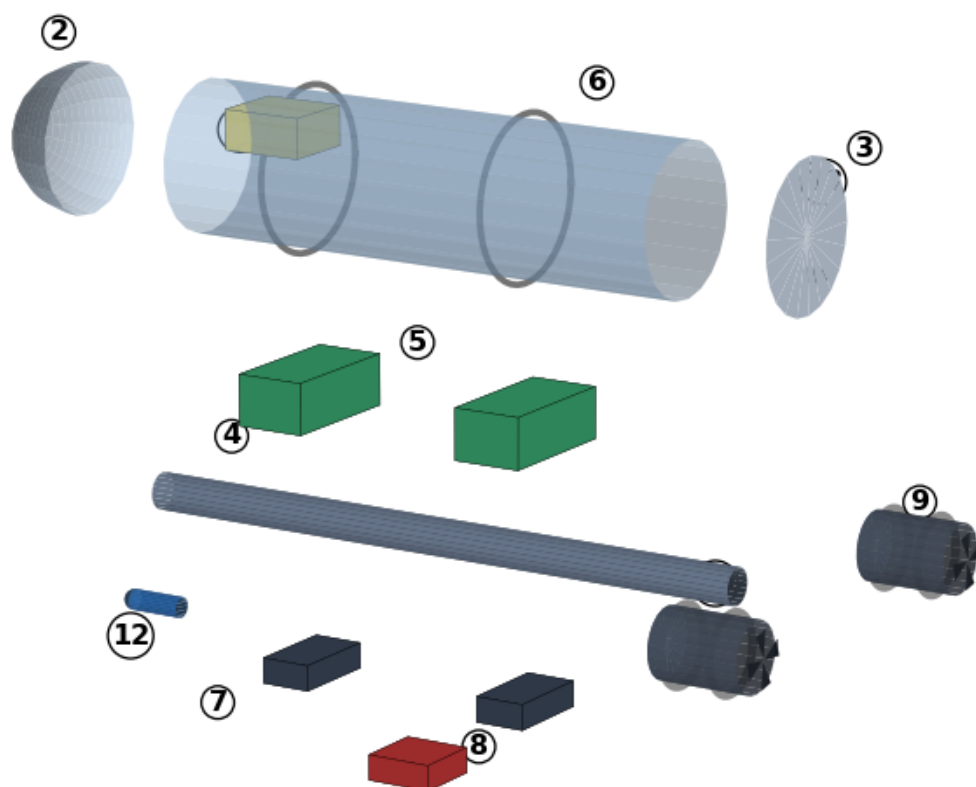


FIG B · Boat exploded

FIG B — Boat exploded

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External parts key

#	Part	Key facts
1	Pressure hull 130 OD × 400 cast acrylic	5 mm wall; never drilled; rides in saddles
2	Acrylic dome + flange (camera end)	lens at centre of curvature; 130 mm
3	Rear cap (DIY 10 mm acrylic on flange)	7 pots + PG7 gland + M6 vent; MS5837 potted (30)
4	Frame rail 32 mm pipe	the backbone — everything clamps to it; drill freely
5	EVA yoga-block saddles ×2	knife-cut 130 arc; positioned over flange zones
6	150 mm jubilee straps ×2	loop hull+saddle+rail; EVA pad under band
7	Fixed lead gym-stacks ×2 (~1 kg ea)	M6 posts + wing nut + nyloc; slotted plates; LOW
8	Drop weight 1–1.5 kg + bridle	continuous loop, ONE burn point, at CG; level release
9	Thruster pods ×2 (60.3 mm PVC)	390 motor + X-mount + shroud; CW/CCW props; couplers 2.3→4
10	P-clips Ø60 ×4 (2 per pod)	rubber-lined; M5+nyloc into drilled rail; axes ■ hull
11	Lamp (headlamp housing, greased seals)	dummy-cell fed from pack; W dimmable / R beacon; on top fwd
12	Speed sensor YF-TM02 + printed intake horn	ram-flow log; low on frame, clear water, serviceable
13	Tether: flat Cat5e + paracord member	cord ties to rail FIRST; PG7 gland; strain never on cable
14	External cabling	zip ties through drilled rail holes every 100–150 mm; service loops; chafe sleeves

FIG C · Sled assembled — inside the hull

FIG C — Sled assembled

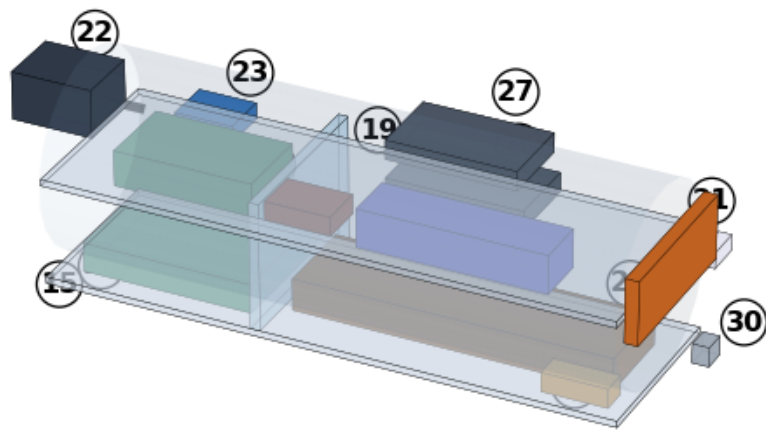
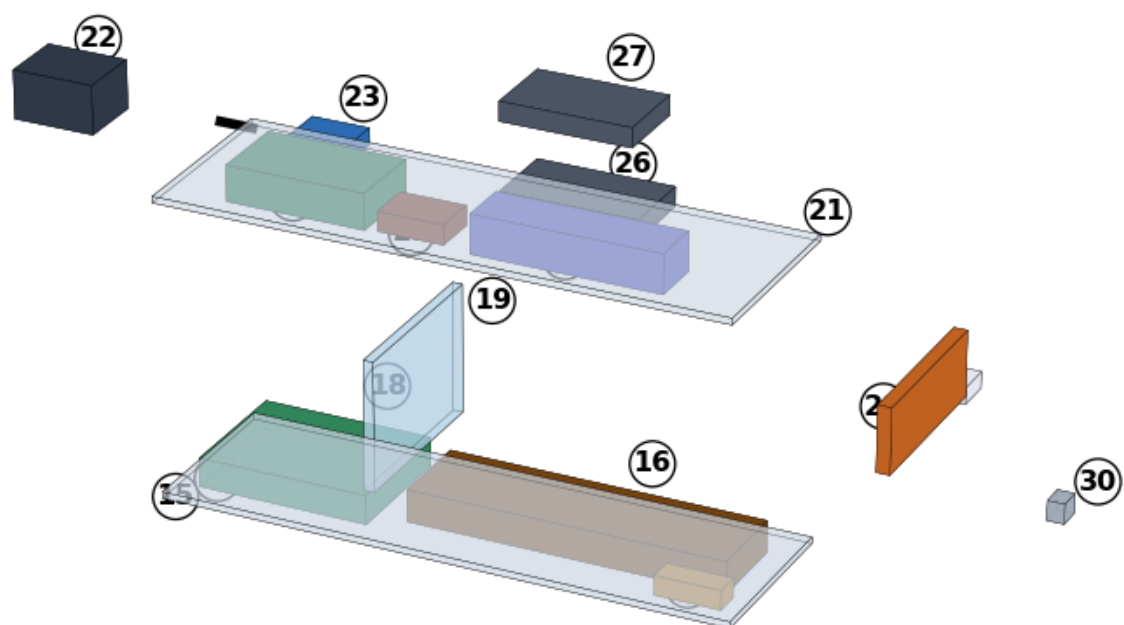


FIG D · Sled exploded

FIG D — Sled exploded



Internal parts key

#	Part	Key facts
15	Floor deck (2 plates solvent-welded, doubler at seam)	heavy & wet layer; sits ~35–40 mm below centreline
16	Battery 3S3P + BMS on rear face + NTC	strapped; rear-biased; seam not under strap anchors
17	10 A main fuse + 50 A isolator	immediately at pack output — minimum unfused wire
18	Ballast bag 500 ml in printed drip tray	velcro; at CG; neck-up self-purging; tube rises to pump
19	Acrylic partition rib	spill wall + Ø119 support rib + seam doubler; vent notch
20	Leak sensors ×3	FWD in tray · MID battery low pt · AFT on hull at cap
21	Mid deck	brains layer at bore centreline; boards on brass standoffs
22	Camera on cantilever arm	lens at hull centre; bench-set tilt on slotted pivots
23	BNO085 IMU	nose end — max distance from motors, lead, pump
24	Raspberry Pi	wlan0→camera AP · eth0→tether · GPIO→DRV8871s only
25	ESP32 brainstem	USB to Pi (tied); owns I ² C, leaks, NTC, flow, lamp, pump, burn interlock
26	Driver board (perfboard)	2× DRV8871 + 3× IRLZ44N + 200R gates + 10k pulldowns
27	Power board (mezzanine, 20 mm standoffs above 26)	LM2596→8 V · 5 V buck · INA219 ×2 · rail fuses; LM2596 fins up
28	Peristaltic pump + inline flow sensor	slow ballast mover; tube → bag below / cap pot aft
29	Connector bus bar (10 mm acrylic, aft face)	XT60 ×3 colour-coded (charge top corner) · JST backplane strip · RJ45 keystone; all plugs point aft — fr
30	MS5837-30BA at rear cap	gel face potted to water; I ² C forward to ESP32 via panel JST

Assembly order: floor deck build (16–20) → mid deck build (22–28) → bus bar (29) → weld decks to ribs → slide-in test with cap off → connect 7 plugs + charge port → cap on → vent-vac test → bathtub ceremony. Teardown is the reverse and must take under five minutes; if it doesn't, move whatever blocked you.